NO.5 i-stop FUNCTION OPERATES UNDER NO ENGINE-STOP CONDITIONS [SKYACTIV-D 1.5]

id1103q4001000

5	i-stop FUNCTION OPERATES UNDER NO ENGINE-STOP CONDITIONS
DESCRIPTION	• i-stop function operates even though system is turned off by pressing i-stop OFF switch. • i-stop function operates even though vehicle conditions are for non-operation. • i-stop function operates frequently while parking vehicle.

5	i-stop FUNCTION OPERATES UNDER NO ENGINE-STOP CONDITIONS
POSSIBLE CAUSE	**i-stop warning light (amber) false illumination** • i-stop warning light (amber) illumination circuit malfunction in instrument cluster • CAN communication line malfunction between instrument cluster and PCM **False detection of i-stop function operation conditions** • False detection of brake pedal depressed — Brake switch No.1 malfunction — Short to power supply in wiring harness between brake switch terminal D and PCM terminal 2AA — Brake fluid pressure sensor (built-into DSC HU/CM) malfunction • False detection of ATF temperature within operable range (**20 to 120 °C {68 to 248 °F}**) (ATX) — TFT sensor malfunction • False detection of external vehicle temperature within operable range (**−10 to 50 °C {14 to 122 °F}**) — Ambient temperature sensor malfunction (sensor specific malfunction) — Short or open circuit in wiring harness between ambient temperature sensor terminal A and PCM terminal 2AG — Open circuit in wiring harness between ambient temperature sensor terminal B and PCM terminal 2AT • Climate control unit falsely detects that internal vehicle temperature is within operation range. (with full-auto air conditioner) — Cabin temperature sensor malfunction (sensor specific or motor malfunction) — Short or open circuit in wiring harness between cabin temperature sensor terminal A and climate control unit terminal F — Open circuit in wiring harness between cabin temperature sensor terminal B and climate control unit terminal E • Falsely detects that climate control unit detects that driver-side air mix door is not in MAX HOT or MAX COLD position even though driver-side air mix door is in MAX HOT or MAX COLD position (with full-auto air conditioner) — Driver-side air mix actuator malfunction — Driver-side air mix actuator position sensor malfunction — Driver-side air mix door or linkage stuck • False detection of vehicle not being parked — False detection of steering wheel rotation speed • Steering angle sensor initialization malfunction • EPS control module malfunction • Falsely detects that vehicle is under safety condition — False detection of closed bonnet • Bonnet latch switch malfunction (stuck closed) • Short to ground in wiring harness between bonnet latch switch terminal A and body control module (BCM) terminal 3W — False detection of closed door, trunk (4SD)/liftgate (5HB) • Door latch switch malfunction • Trunk lid latch switch (4SD) • Liftgate latch switch malfunction (5HB) • Short to ground in wiring harness between door latch switch and body control module (BCM) • Open circuit in wiring harness between trunk lid latch switch and body control module (BCM) (4SD) • Open circuit in wiring harness between liftgate latch switch and body control module (BCM) (5HB) — False detection of fastened driver seat belt • Driver-side buckle switch malfunction • Open circuit in wiring harness between driver-side buckle switch terminal 5B and SAS control module terminal 2U — False detection of inclination angle (false detection of **7 % or less**) • Low-G (XY) sensor (built-into SAS control module) malfunction (In this case, the SAS control module records DTCs C0061:29 and C0062:29.) — False detection of low power brake unit load • Power brake unit vacuum sensor malfunction • Short or open circuit in wiring harness between power brake unit vacuum sensor terminal A and PCM terminal 2AK • Short or open circuit in wiring harness between power brake unit vacuum sensor terminal B and PCM terminal 2BD • Short or open circuit in wiring harness between power brake unit vacuum sensor terminal C and PCM terminal 2BC

Diagnostic Procedure

STEP	INSPECTION	RESULTS	ACTION
1	**VERIFY DTC** • Retrieve the PCM, TCM, body control module (BCM), DSC HU/CM, SAS control module, instrument cluster and climate control unit DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].) (See DTC INSPECTION [BCM].) (See DTC INSPECTION [DSC HU/CM].) (See DTC INSPECTION [STANDARD DEPLOYMENT CONTROL SYSTEM].) (See DTC INSPECTION [TWO-STEP DEPLOYMENT CONTROL SYSTEM].) (See DTC INSPECTION [INSTRUMENT CLUSTER].) (See DTC DISPLAY [FULL-AUTO AIR CONDITIONER].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].) (See ON-BOARD DIAGNOSTIC SYSTEM DTC TABLE [TCM (EW6A-EL)].) (See DTC TABLE [BCM].) (See DTC TABLE [DSC HU/CM].) (See DTC TABLE [STANDARD DEPLOYMENT CONTROL SYSTEM].) (See DTC TABLE [TWO-STEP DEPLOYMENT CONTROL SYSTEM].) (See DTC TABLE [INSTRUMENT CLUSTER].) (See DTC TABLE [FULL-AUTO AIR CONDITIONER].)
		No	Go to the next step.
2*	**DETERMINE IF MALFUNCTION CAUSE IS AMBIENT TEMPERATURE SENSOR SIGNAL OR OTHER** • Switch the ignition ON (engine off). • Compare the ambient temperature sensor on the LCD with the actual ambient temperature. • Does the ambient temperature on the LCD correspond to the actual ambient temperature?	Yes	Go to Step 4.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
3	**INSPECT AMBIENT TEMPERATURE SENSOR** • Inspect the ambient temperature sensor. (See AMBIENT TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is there any malfunction?	Yes	Replace the ambient temperature sensor. (See AMBIENT TEMPERATURE SENSOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].)
		No	Inspect the following: • Short or open circuit in wiring harness between ambient temperature sensor terminal A and PCM terminal 2AG **If there is any malfunction:** — Refer to the wiring diagram and verify whether or not there is a common connector between ambient temperature sensor terminal A and PCM terminal 2AG. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short or open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short or open circuit. • Open circuit in wiring harness between ambient temperature sensor terminal B and PCM terminal 2AT **If there is any malfunction:** — Refer to the wiring diagram and verify whether or not there is a common connector between ambient temperature sensor terminal B and PCM terminal 2AT. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit.
4*	**DETERMINE IF MALFUNCTION CAUSE IS DOOR LATCH SWITCH AND TRUNK LID LATCH SWITCH (4SD)/LIFTGATE LATCH SWITCH (5HB) SIGNAL OR OTHER** • Switch the ignition ON (engine off). • Access the following body control module (BCM) PIDs using the M-MDS: (See PID/DATA MONITOR INSPECTION [BCM].) — TRUNK — DOOR_D — DOOR_ALL • Are the PID values congruent with the opening and closing of the doors and trunk (4SD)/liftgate (5HB)? (See PID/DATA MONITOR TABLE [BCM].)	Yes	Go to Step 6.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
5	**INSPECT DOOR LATCH SWITCH AND TRUNK LID LATCH SWITCH (4SD)/LIFTGATE LATCH SWITCH (5HB)** • Inspect the PID-related switch in which the malfunction occurred in Step 4. (See TRUNK LID LATCH SWITCH INSPECTION.) (See LIFTGATE LATCH SWITCH INSPECTION.) (See FRONT DOOR LATCH SWITCH INSPECTION.) (See REAR DOOR LATCH SWITCH INSPECTION.) • Is there any malfunction?	Yes	Replace the applicable switch. (See TRUNK LID LATCH AND RELEASE ACTUATOR REMOVAL/INSTALLATION.) (See LIFTGATE LATCH AND LOCK ACTUATOR REMOVAL/INSTALLATION.) (See FRONT DOOR LATCH AND LOCK ACTUATOR REMOVAL/INSTALLATION.) (See REAR DOOR LATCH AND LOCK ACTUATOR REMOVAL/INSTALLATION.)
		No	Inspect the following in which the malfunction occurred in Step 4: • Short to ground in wiring harness between door latch switch and body control module (BCM) **If there is any malfunction:** — Refer to the wiring diagram and verify whether or not there is a common connector between door latch switch and body control module (BCM). **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Open circuit in wiring harness between trunk lid latch switch (4SD)/liftgate latch switch (5HB) and body control module (BCM) **If there is any malfunction:** — Refer to the wiring diagram and verify whether or not there is a common connector between trunk lid latch switch (4SD)/liftgate latch switch (5HB) and body control module (BCM). **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 20.
6*	**DETERMINE IF MALFUNCTION CAUSE IS DRIVER-SIDE BUCKLE SWITCH SIGNAL OR OTHER** • Switch the ignition ON (engine off). • Access the SAS control module PID SEAT_B_D using the M-MDS. (See PID/DATA MONITOR INSPECTION [STANDARD DEPLOYMENT CONTROL SYSTEM].) (See PID/DATA MONITOR INSPECTION [TWO-STEP DEPLOYMENT CONTROL SYSTEM].) • Is the SEAT_B_D PID value congruent with the seat belt condition? (See PID/DATA MONITOR TABLE [STANDARD DEPLOYMENT CONTROL SYSTEM].) (See PID/DATA MONITOR TABLE [TWO-STEP DEPLOYMENT CONTROL SYSTEM].)	Yes	Vehicles with manual air conditioner: • Go to Step 10. Vehicles with full-auto air conditioner: • Go to Step 8.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
7	**INSPECT DRIVER-SIDE BUCKLE SWITCH** • Inspect the driver-side buckle switch. (See BUCKLE SWITCH INSPECTION.) • Is there any malfunction?	Yes	Replace the driver-side buckle switch. (See FRONT BUCKLE REMOVAL/INSTALLATION.)
		No	Repair or replace the wiring harness between driver-side buckle switch terminal 5B and SAS control module terminal 2U for a possible open circuit.
8	**DETERMINE IF MALFUNCTION CAUSE IS CABIN TEMPERATURE SENSOR SIGNAL OR OTHER** • Access the climate control unit PID INC_TMP_SEN using the M-MDS. (See PID/DATA MONITOR DISPLAY [FULL-AUTO AIR CONDITIONER].) • Does the INC_TMP_SEN PID value indicate the actual cabin temperature of the vehicle?	Yes	Go to Step 10.
		No	Go to the next step.
9	**INSPECT CABIN TEMPERATURE SENSOR** • Inspect the cabin temperature sensor. (See CABIN TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is there any malfunction?	Yes	Replace the cabin temperature sensor. (See CABIN TEMPERATURE SENSOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].) Go to Step 20.
		No	Inspect the wiring harness between the following terminals for a short or open circuit: • Cabin temperature sensor terminal A—Climate control unit terminal F • Cabin temperature sensor terminal B—Climate control unit terminal E **If there is any malfunction:** — Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Cabin temperature sensor terminal A—Climate control unit terminal F • Cabin temperature sensor terminal B—Climate control unit terminal E **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short or open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short or open circuit. Go to Step 20.
10*	**DETERMINE IF MALFUNCTION CAUSE IS BONNET LATCH SWITCH SIGNAL OR OTHER** • Switch the ignition ON (engine off). • Access the body control module (BCM) PID HOOD using the M-MDS. (See PID/DATA MONITOR INSPECTION [BCM].) • Is the HOOD PID value normal? (See PID/DATA MONITOR TABLE [BCM].)	Yes	Go to Step 12.
		No	Go to the next step.
11	**INSPECT BONNET LATCH SWITCH** • Inspect the bonnet latch switch. (See BONNET LATCH SWITCH INSPECTION.) • Is there any malfunction?	Yes	Replace the bonnet latch switch. (See BONNET LATCH REMOVAL/INSTALLATION.) Go to Step 20.
		No	Repair or replace the wiring harness between bonnet latch switch terminal A and body control module (BCM) terminal 3W for a possible short to ground. Go to Step 20.

STEP	INSPECTION	RESULTS	ACTION
12	**DETERMINE IF MALFUNCTION IS CAUSED BY STEERING ANGLE (ESTIMATED ABSOLUTE ANGLE) SIGNAL ERROR** • Start the engine and idle it. • Using the M-MDS, display EPS control module PID STR_AB_EST. (See PID/DATA MONITOR INSPECTION [EPS CONTROL MODULE].) • Are the monitoring values normal?	Yes	Go to Step 14.
		No	Go to the next step.
13	**INSPECT EPS CONTROL MODULE FOR MALFUNCTION** • Inspect the EPS control module. (See EPS CONTROL MODULE INSPECTION.) • Is the EPS control module normal?	Yes	Perform the following procedure: 1. Switch the ignition to OFF, and maintain this condition for approx. 3 s. 2. Set the vehicle wheels in the straight-ahead position. 3. Switch the ignition ON (engine off or on). 4. Using the M-MDS, display EPS control module PID STR_AB_EST. (See PID/DATA MONITOR INSPECTION [EPS CONTROL MODULE].) • If the STR_AB_EST value is normal, go to Step 20. (Because the steering angle (estimated absolute angle) has returned to normal) • If the STR_AB_EST value is not normal, replace the EPS control module, then go to Step 20. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
		No	Replace the EPS control module, then go to Step 20. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
14	**DETERMINE IF MALFUNCTION CAUSED BY BRAKE OPERATION SIGNAL ERROR** • Switch the ignition ON (engine off). • Access the PCM PID BOO using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Does the BOO PID value change according to the brake pedal operation? (See PCM INSPECTION [SKYACTIV-D 1.5].)	Yes	Go to Step 16.
		No	Go to the next step.
15	**INSPECT BRAKE SWITCH** • Inspect the brake switch. (See BRAKE SWITCH INSPECTION.) • Is there any malfunction?	Yes	Replace the brake switch. (See BRAKE PEDAL REMOVAL/INSTALLATION [L.H.D.].) (See BRAKE PEDAL REMOVAL/INSTALLATION [R.H.D.].) Go to Step 20.
		No	Open circuit in wiring harness between brake switch terminal D and PCM terminal 2AA. • Repair or replace the suspected wiring harness. Go to Step 20.
16	**DETERMINE IF MALFUNCTION CAUSE IS POWER BRAKE UNIT VACUUM SENSOR SIGNAL OR OTHER** • Turn off the i-stop system with i-stop OFF switch. • Start the engine and run it is idling. • Switch the ignition off. • Switch the ignition ON (engine off). • Access the PCM PID BBP using the M-MDS while the brake pedal has been depressed several times. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Does the monitor value decrease every time the brake pedal is depressed?	Yes	Vehicles with manual air conditioner: • Go to Step 20. Vehicles with full-auto air conditioner: • Go to Step 18.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
17	**INSPECT POWER BRAKE UNIT VACUUM SENSOR** • Inspect the power brake unit vacuum sensor. (See POWER BRAKE UNIT VACUUM SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the power brake unit vacuum sensor. (See POWER BRAKE UNIT REMOVAL/INSTALLATION [L.H.D.].) (See POWER BRAKE UNIT REMOVAL/INSTALLATION [R.H.D. (SKYACTIV-D 1.5)].)
		No	Inspect the wiring harness between the following terminals for a short or open circuit: • Power brake unit vacuum sensor terminal A—PCM terminal 2AK • Power brake unit vacuum sensor terminal B—PCM terminal 2BD • Power brake unit vacuum sensor terminal C—PCM terminal 2BC **If there is any malfunction:** — Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Power brake unit vacuum sensor terminal A—PCM terminal 2AK • Power brake unit vacuum sensor terminal B—PCM terminal 2BD • Power brake unit vacuum sensor terminal C—PCM terminal 2BC **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short or open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short or open circuit. **If there is no malfunction:** — Replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
18	**DETERMINE IF MALFUNCTION CAUSE IS DRIVER-SIDE AIR MIX ACTUATOR SIGNAL OR OTHER** • Measure the voltage at the following terminal (wiring harness-side) when the driver-side temperature setting is MAX HOT and MAX COLD. — Climate control unit terminal G • Is the voltage normal? (See CLIMATE CONTROL UNIT INSPECTION [FULL-AUTO AIR CONDITIONER].)	Yes	Repeat the inspection from Step 1. • If the malfunction is not resolved, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 20.
		No	Go to the next step.
19	**INSPECT DRIVER-SIDE AIR MIX ACTUATOR** • Inspect the driver-side air mix actuator. (See AIR MIX ACTUATOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is there any malfunction?	Yes	Replace the driver-side air mix actuator. (See AIR MIX ACTUATOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].)
		No	Inspect the air mix actuator and linkage for sticking. (See A/C UNIT DISASSEMBLY/ASSEMBLY.) • If there is any malfunction: — Repair or replace the malfunctioning part according to the inspection results.
20	Verify the test results. • If normal, return to the diagnostic index to service any additional symptoms. (See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].) • If a malfunction remains, inspect the related Service Information and perform the repair or diagnosis. — If the vehicle is repaired, troubleshooting is completed. — If the vehicle is not repaired or additional diagnostic information is not available, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)		